

What Did Ten Points of Mean Recall Buy?

Separating Group-Level from Case-Level Gains in Scene Graph Generation

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Two frozen models, one cell: relabelling within ϕ removes case identity and nothing else

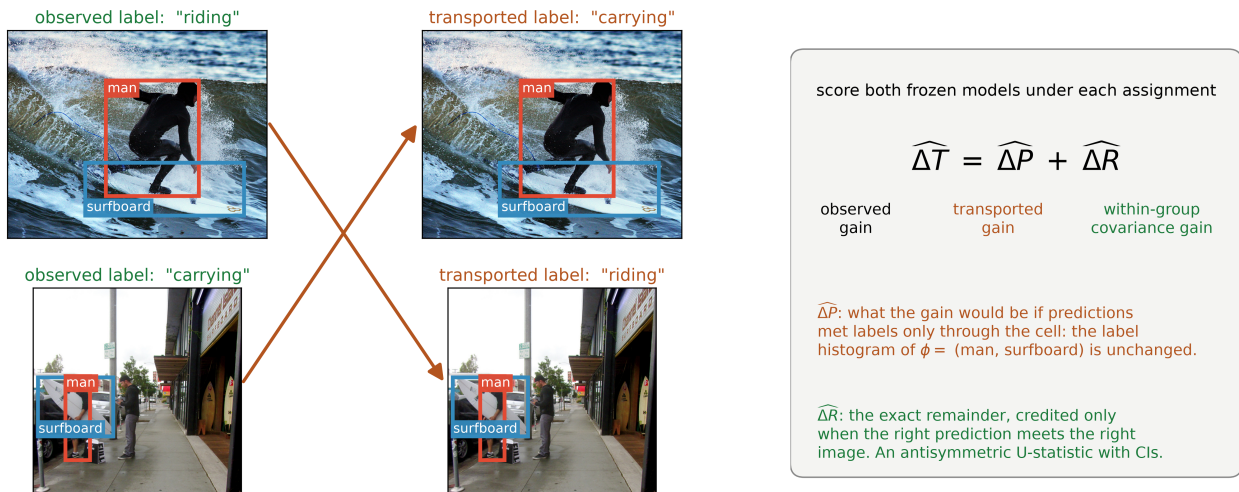


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair’s predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\widehat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\widehat{\Delta R}$ is the part that needs the right label on the right image.

Abstract

001 Debiasing methods for scene graph generation are judged
 002 by mean recall, and one of the best known, Total Direct Ef-
 003 fect (TDE), raises it on Visual Genome from 14.6 to 24.8.
 004 What did those ten points buy? Mean recall cannot say. A
 005 model raises it either by predicting rare predicates more of-
 006 ten within each subject–object pair, or by recognising which
 007 individual images actually show them, and the metric cred-
 008 its both alike. We give an exact way to tell them apart.
 009 Scoring each prediction against the label of a different im-
 010 age from the same subject–object group keeps the group’s
 011 predicate frequencies but breaks the link between image and
 012 label; this splits any proper-score difference between two
 013 frozen models into a group-level part and a case-level co-

variance, with no fitted baseline or tuning parameter, in
 $O(NK)$ time, with image-clustered confidence intervals.
 On TDE’s released checkpoints with calibration matched,
 the change is almost entirely group-level: the case-level
 part is at most a twelfth of it under either proper score, and
 under the quadratic score indistinguishable from zero. The
 ten-point gain is therefore not evidence that the debiased
 model recognises individual relations better. The converse
 also occurs: a CLIP model reading image crops, ranked last
 on accuracy and on the proper score, discriminates individ-
 ual relations significantly better than the geometry model
 ranked above it.

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1. Introduction

Scene graph generation is evaluated on a benchmark whose labels are unusually predictable from something every model sees. Given the ordered object pair (man, surfboard), the predicate on Visual Genome [14] is very often riding, and a lookup table of training frequencies is famously competitive [33]. Recall@ K rewards models that echo this prior, so the field now leads with mean recall, which averages recall over predicates and so rewards getting the rare ones [29]. Debiasing methods are now judged by how far they raise it. Total Direct Effect (TDE) [29], among the most cited of them, raises mean recall@50 of a MOTIFS model from 14.6 to 24.8 on the canonical VG150 split, in our evaluation of the released checkpoint with the authors’ own code.

What did those ten points buy? There are two ways to raise mean recall, and the metric cannot tell them apart. A model can move probability towards rare predicates *within each subject–object pair* — say carrying more often for (man, surfboard) — without becoming any better at deciding *which* images show a man carrying rather than riding. Or it can learn to make that decision: put more probability on the correct predicate for the individual image, beyond what the object pair implies. The first is a change in the model’s group-level predictions; only the second is better recognition. Every existing remedy — frequency-stratified recall [18], zero-shot recall [20], rarity-stratified accuracy [13] — slices one model’s performance differently. None attributes the gain *between* two models to one route or the other.

An exact split. We separate the two routes with one operation (Fig. 1). Take two frozen models and a test set grouped by a discrete feature both models observe, here the subject–object pair. Score each model’s prediction for one relation against the label of a *different* relation in the same group. This relabelling preserves each group’s predicate frequencies exactly, so any part of the score gain earned at the level of the group survives it; it destroys the correspondence between an image and its own label, so any part earned by getting individual cases right does not. The observed gain therefore splits exactly into a *group-level* part $\widehat{\Delta P}$ and a *case-level* remainder $\widehat{\Delta R}$, which is a within-group covariance between the change in the models’ predictions and the label (Theorem 1). The split holds in the observed sample for any proper score, needs no fitted reference model, held-out fold or tuning parameter, costs $O(NK)$ on cached predictions, and comes with image-clustered intervals.

What it finds. On the two models of TDE’s own released checkpoint, evaluated with the authors’ code, and with each model’s confidence matched on held-out images,

the proper-score change is almost entirely group-level: the case-level part is at most a twelfth of it under either the quadratic or the log score, and under the quadratic score it is -0.00006 with an interval straddling zero (Sec. 5). So the ten points are not evidence that the debiased model recognises individual relations better: the proper-score change behind them is a redistribution of probability across predicates within each pair. This describes what TDE does rather than indicting it — subtracting a context-only prediction is a group-level intervention by design — but it is not what a ten-point gain is usually read as meaning. The converse happens too. A model reading image crops through a frozen CLIP encoder [24], which every aggregate measure ranks last, gains $+0.00641$ of case-level covariance over the geometry-based model ranked above it, with an interval excluding zero (Sec. 6): its deficit is entirely group-level, and it is the better recogniser.

Contributions.

- An audit of a published debiasing method’s released checkpoints showing that its ten-point mean-recall gain carries no detectable case-level improvement under the quadratic score, and at most a twelfth of the change under the log score, and a reversal in which the model ranked last is the best case-level discriminator.
- An exact, model-free decomposition of the score gain between two frozen models into group-level and case-level parts, valid for every Bregman score, computed in one pass over cached predictions, with cluster-robust inference.
- Validation that the case-level part is exactly zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects — together with a protocol for the one confound it does not separate.

What is not claimed. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model’s confidence moves score between the two parts while changing no ranking — which is why every comparison here is confidence-matched. And the audit’s headline holds at the configuration we argue for, not at every configuration; Tab. 1 reports all five. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation. Scene graph generation was posed as structured prediction over a whole image, solved by message passing between

object and relation nodes [31]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [33]. Mean recall became the headline metric, and the causal TDE intervention was introduced, to reduce that dependence [29]; a long line of debiasing work has followed [15, 21], and the metrics themselves have been shown to reward pathologies [18], a diagnosis that recurs in open-vocabulary settings [17]. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change between train and test [1], GQA-OOD stratifies accuracy by answer rarity within question groups [13], and multimodal models exploit non-visual shortcuts [9, 34]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 22]. All of these characterise a *single* model. We attribute the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [6, 23], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [32], which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [28], but not for a contrast between two models scored on the same data.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [8] and its conditional form [10]; at a single group our total-gain interval is a clustered Diebold–Mariano test (Corollary 3), so the decomposition adds a split to what those tests already report. DeLong *et al.* [7] compare correlated ROC areas with a paired U-statistic, the same device we use for a group-conditional score gain. The case-level term is an order-two U-statistic [12], with a Hájek projection supplying its influence function [27].

3. Splitting a score gain

Setup. Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K, \quad (1)$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c , let $\Delta T_c = \mathbb{E}[H_X(Y) | C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\begin{aligned} \Delta T_c &= \Delta P_c + \Delta R_c, \\ \Delta R_c &= \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \end{aligned} \quad (2)$$

For the quadratic score, with $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP , which is why we call it the group-level part. It is *not* simply “prior fit”: under the quadratic score ΔP_c is the improvement in how close the model’s mean forecast sits to the cell’s label frequencies *minus* the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [32] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y -th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. One cell, four cases, three labelled 1 and one labelled 2; the old model predicts $(\frac{1}{2}, \frac{1}{2})$ for all four. Model A predicts $(\frac{3}{4}, \frac{1}{4})$ for every case — the cell’s frequencies. Model B predicts $(\frac{3}{4}, \frac{1}{4})$ for the three label-1 cases and $(\frac{1}{4}, \frac{3}{4})$ for the label-2 case: the same distance from the old model, but pointed at each case’s own label. Under the quadratic score A gains $+\frac{1}{8}$, all of it group-level, and B gains $+\frac{3}{8}$ with a case-level part of $+\frac{1}{2}$. Now let deployment see this cell’s labels in proportions $(\frac{1}{4}, \frac{3}{4})$ instead, a label shift. Reweighting each case by $p'(y)/p(y)$ sends A to $-\frac{3}{8}$ — worse than the model it

replaced, with not one prediction changed — and leaves B at $+\frac{3}{8}$. The split said in advance which gain was exposed: a group-level gain is contingent on the evaluation set’s label frequencies matching deployment, and a case-level gain is not. That is the operational reason to want them apart. (Appendix C works the example through.)

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_i H_i(y_i), \quad \widehat{\Delta P}_c = \frac{1}{n_c(n_c-1)} \sum_{i \neq j} H_i(y_j), \quad (4)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j} A_{ij}, \quad (5)$$

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (6)$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [12] for ΔR_c (Theorem 3). Transporting a case only to the *other* cases’ labels matters: fitting the cell’s label frequencies on all n_c cases, the natural plug-in, shrinks every cell’s case-level gain by exactly $(n_c - 1)/n_c$, halving it at $n_c = 2$ (Proposition 4). With m_{cy} the count of label y in cell c , case i ’s transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \quad (7)$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. With $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$, the Hájek influence contribution of case i is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}.$$

We sum $\widehat{\psi}_i$ within each image and use the one-way cluster-robust sandwich variance, since relations from one photograph are not independent. The estimator is asymptotically normal under bounded scores and non-dominating clusters (Theorem 4); at a single cell, the total-gain interval is a clustered Diebold–Mariano test [8] (Corollary 3). A randomization test that cyclically shifts labels within every cell is a secondary, relation-level diagnostic.

Confidence matching. One property of the split must be part of the procedure rather than a caveat. Because ΔR is a covariance between probability *movements* and labels, rescaling a fixed model’s confidence moves score between the two parts while changing no ranking, no top-1 decision and no information about any case (Sec. 4 measures how

much). Two released checkpoints that differ in calibration therefore cannot be compared on their raw outputs. Whenever compared models differ visibly in confidence we split the images at random, fit each model a single temperature by maximum likelihood on one half, and run the split on the other half with each model’s own temperature applied. The protocol uses no label from the audited half.

Choosing the grouping. The split is conditional on ϕ . Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 2), so the finest grouping that the benchmark’s construction defends is the conservative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition, and Proposition 3 bounds it by a single elicited correlation, in the spirit of the robustness value of Cinelli and Hazlett [4].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. We draw 24 cells over five labels with Dirichlet label distributions, take the old model to be exactly the cell prior, and let the new one interpolate towards an oracle. The estimated case-level gain grows linearly with the injected signal. Under a null that perturbs the prior predictions and reassigns them within cells, the randomization test rejects in 0.030 of 200 datasets at level 0.05, with mean estimate -8.97×10^{-5} . For the interval we use on real data, we simulate relations clustered into images with a shared per-image effect and Zipf-distributed cell sizes, most cells holding two cases: the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for an interval that treats relations as independent.

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of -0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to -0.0003 (sd 0.0017). That last design is why the protocol is part of the procedure.

Controlled predictors on VG150. We train predicate classifiers on the VG150 PredCls split whose inputs we control: a frequency table that sees only the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations, 99.0%, in 6,346 identified class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is algebraic, not empirical — both models are functions of the class pair alone, so Theorem 1 forces it — and it is the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry then yields a case-level gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score isolates. Against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part deteriorates by 0.00554: geometry made the model a better recogniser and a slightly worse fit to the cell frequencies at once.

5. What TDE’s ten points bought

Tang *et al.* [29] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE), which subtracts a context-only counterfactual prediction at inference. We evaluate both with the authors’ own code on the canonical VG150 test split, so the two prediction sets are aligned relation for relation. The reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $mR@50 = 0.2476$ and raising zero-shot recall to $zR@50 = 0.1432$, while top-1 predicate accuracy falls from 0.6981 to 0.5577. The audit covers 183,639 relations over 26,446 images in 7,127 subject–object cells; 98.9% of the relations fall in the 5,044 cells holding at least two, and so are identified.

The released pair must be confidence-matched. The two models differ sharply in calibration: the temperatures fitted on held-out images are $T^* = 2.50$ for the baseline and 0.92 for TDE. Scored as they ship, they give a significant case-level *loss* for TDE (−0.01355, Tab. 1), but Sec. 4 showed that a calibration difference alone manufactures exactly that kind of loss. We therefore read the matched rows.

Result. Once each model carries its own held-out temperature, TDE’s quadratic-score change is −0.18264, of which −0.18258 is group-level. The case-level part is

Table 1. **TDE against its MOTIFS baseline** on VG150 PredCls, released checkpoints. Raw rows use every identified relation; matched rows fit each model’s temperature on a held-out half of the images and audit the other half. The interval on $\widehat{\Delta R}$ is image-clustered. Our reading is the matched quadratic row; all five are reported because the answer depends on the configuration.

	Score	ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ [95% CI]
raw	quad.	pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	pair	+0.03277	+0.15897	−0.12619 [−0.13093, −0.12145]
raw	quad.	subject	−0.11536	+0.10175	−0.21711
matched	quad.	pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
matched	log	pair	−0.54760	−0.59157	+0.04396 [+0.04079, +0.04714]

$\widehat{\Delta R} = -0.00006$, interval [−0.00120, +0.00109], randomization $p = .555$: indistinguishable from zero (Fig. 2). Under the log score the case-level part is small and positive, +0.04396 with an interval excluding zero, against a group-level change of −0.59157. The two proper scores thus disagree on whether any case-level change survives matching, and agree that whichever is right it is at most a twelfth the size of the total. Both are well-posed: each Bregman score defines its own exact estimand (Theorem 2), so choosing between them is a declaration and not an error to argue away. What both support is that TDE’s change to the predictive distribution is overwhelmingly group-level.

What that does and does not license. Mean recall rose ten points; the matched case-level gain is −0.00006. Those are different units and we supply no bridge between them: it would be wrong to say ten points bought 0.00006 of evidence, and equally wrong to say the mean-recall gain *is* group-level, since the group-level part of the proper score fell. What the audit supports is a conjunction. The headline metric moved ten points, and the proper-score difference behind it lives almost entirely in how the model distributes probability across predicates within each object pair. A reader may conclude that the ten points are not evidence of better recognition of individual relations. This characterises TDE rather than indicting it: subtracting a context-only prediction is a group-level intervention by construction, and the split confirms that it acts as designed. What it questions is the habit of reading a mean-recall gain as better recognition.

The answer depends on the configuration. The five rows give case-level gains of −0.21711, −0.12619, −0.01355, −0.00006 and +0.04396. At two of the five, “overwhelmingly group-level” is false: the raw log row has $|\widehat{\Delta R}| = 0.126$ against $|\widehat{\Delta P}| = 0.159$, and the subject-

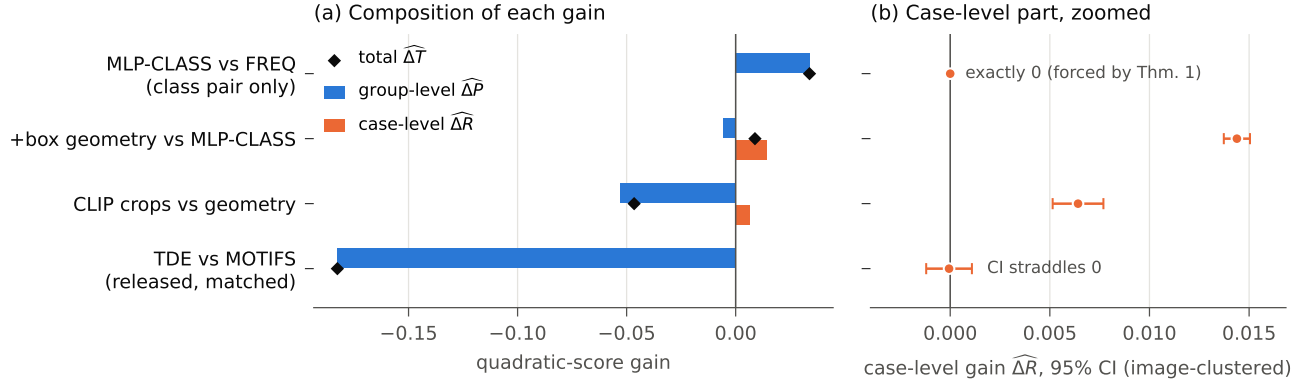


Figure 2. **Where each gain in this paper lives**, under the quadratic score. (a) Every comparison’s total gain $\widehat{\Delta T}$ (diamond) split into its group-level part $\widehat{\Delta P}$ and case-level part $\widehat{\Delta R}$; the two parts sum to the total exactly. (b) The case-level part alone with its image-clustered 95% interval. Two models built from the class pair alone differ by a purely group-level gain, as Theorem 1 forces. Box geometry and CLIP crops each buy a significant case-level gain, the CLIP model despite losing on the total. TDE’s large proper-score change is group-level, with a case-level part whose interval straddles zero.

only row has -0.217 dominating $+0.102$. We argue for the matched class-pair row on stated grounds — a proper score punishes miscalibration, so calibration must be held fixed; and coarsening is not neutral (Proposition 2), so the finest grouping the benchmark defends is the conservative one — and report the rest so that a reader who rejects those grounds can see what follows instead.

6. The model every metric ranks last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. We add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [24], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of 100,000 training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for the geometry model, and a quadratic score 0.04655 lower. An evaluator reading either number would discard it. The split says the opposite on the axis that matters. Against the geometry model, the CLIP model loses 0.05296 of group-level score while *gaining* 0.00641 of case-level score (CI $[0.00514, 0.00769]$, $p = .002$): its whole deficit is group-level, and it discriminates between individual relations significantly better (Fig. 2). Against the class-only model, pixels buy 0.01665 of case-level score where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy that stands in for it, and a benchmark that reports

aggregate scores ranks it last anyway.

Two checks support this reading. Post-hoc correction of a model’s label prior should move only the group-level part, and it does: $+0.02401$ group-level against $+0.00111$ case-level. Applied to the CLIP model it halves the aggregate deficit, from -0.04655 to -0.02143 , while the case-level advantage persists ($+0.00752$; $+0.00663$ with both models corrected). And the advantage survives confidence matching: $+0.00467$, CI $[+0.00314, +0.00621]$.

Beyond scene graphs. Nothing in Sec. 3 is specific to relations, and the supplementary applies the identical estimator to long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, with superclasses as the grouping, and to long-tailed text classification. On CIFAR-100-LT two standard re-weighting schedules turn out to have opposite compositions once confidence is matched, reversing what their raw scores suggest about both. The constraint on the grouping bites here: the fine class label cannot serve as a cell, because it leaves every cell label-homogeneous and forces the case-level part to zero by construction — an artefact that reads exactly like a null result.

7. Limitations and conclusion

Limitations. Four bound what the split can say, and we would rather state them than have them found. *It is conditional on the grouping.* The case-level part measures discrimination among relations sharing an object pair; two equally fine partitions of the same cases can disagree completely about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision that the benchmark, not the submitting author, should make before a leaderboard

opens. *It credits any within-cell signal*, including shortcuts nobody recorded, and our own bound on that exposure is not reassuring: for the controlled-predictor pair of Sec. 4, an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain (Appendix I). *It is not invariant to recalibration*, which is why every comparison here is confidence-matched; a rank-based within-cell contrast would be invariant, at the cost of the exact additivity that is the point of the construction. And *it needs probabilities*: most benchmarks archive ranked predictions, and that, not compute, is the obstacle to routine use.

Conclusion. A ten-point mean-recall gain is two different achievements summed together, and the field’s metrics achieve the sum. Scoring each prediction against another case’s label from the same group separates them exactly, with no fitted reference model and with intervals that respect the clustering of relations within images. Applied to a released debiasing method, the split finds a large metric gain carrying no detectable case-level improvement under the quadratic score; applied to a model every aggregate metric ranks last, it finds the best recogniser of the three. We suggest that benchmarks whose labels lean on a feature every model sees report the split beside the headline metric: the declared grouping and its coverage, both signed parts with the case-level interval, and the confidence-matched row. Code and cached predictions to reproduce every number are in the supplementary material.

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